Zakariya Arale

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TECHNICAL SKILLS

Languages: Python, Java, C, C++, HTML, CSS, JavaScript, Bash/Shell scripting, Arm32v7
Frameworks & Libraries: React, Tailwind, Pandas, Scikit-Learn, Matplotlib
Tools & Technologies: Git/GitHub, Linux, VS Code, Jira, Socket Programming, Asynchronous Programming, Android Studio
Concepts & Skills: Object-Oriented Programming (OOP), Data Structures, Algorithms, Machine Learning, SDLC, Scrum, Agile Methodologies, Google Cloud Platform (GCP)

EDUCATION

Honours Bachelor of Science **2024 - Present**
University of Toronto Scarborough

- Software Engineering Co-op + Math Major (GPA: 3.98)

EXPERIENCE

Teacher Assistant - Linear Algebra **Jan 2026 - May 2026**
University of Toronto

- Developed and delivered original lesson plans for weekly tutorials of 30+ students, teaching core concepts in a clear, simple, and concise manner
- Proactively identified common errors and took the initiative to create supplemental resources, ensuring students improved understanding of core concepts
- Mentor students during office hours, guiding them through complex proofs, demonstrating detailed calculation steps, enhancing their mathematical problem solving skills
- Assess and grade coursework, providing constructive feedback to help students improve their responses and deepen their understanding of linear algebra

PROJECTS

MicroC-Complier | Python, Arm32v7, Bash/Shell scripting **Mar 2026 - Apr 2026**

- Engineered an AST-based compiler backend using Python objects to represent C language, translating structured syntax trees into ARM32v7 assembly code
- Implemented static type checking using symbol tables and scope resolution on the AST, detecting semantic errors prior to code generation and improving correctness
- Designed a custom runtime memory model to coordinate how the program stores and retrieves data, supporting function parameters, local variables, and dynamic memory allocation
- Developed Bash scripts to automate build, compilation, and execution on an ARM emulator, improving reliability and development efficiency

SmartAir | Java, XML, Android Studio, Firestore, Scrum, Jira, OOP, SDLC **Nov 2025 - Dec 2025**

- Recognized as one of the top 5 team projects in a class of 200 students and selected by a healthcare agency for use of the application code
- Served as Scrum Master by leading 3 Agile sprint cycles, organizing tasks, and coordinating the team using Jira, resulting in delivery of all project requirements within 3 weeks

- Implemented structured data persistence for respiratory metrics (PEF, PB, triage logs) using Firestore, enabling clinical-grade reporting via automated PDF exports
- Achieved 100% branch test coverage for the authentication module by refactoring it using the MVC design model and implementing comprehensive unit tests with JUnit 5 and Mockito

Async_Battleship | C, Linux, Makefile **Jul 2025 - Sept 2025**

- Engineered an asynchronous gaming server in C using Epoll based I/O multiplexing to support 100+ TCP connections with minimal delays
- Ensured reliable message processing for users with unstable connections by implementing non-blocking reads and buffer shifting to process fragmented TCP commands
- Automated build and execution workflows using Makefile, reducing manual setup time and improving deployment efficiency

AccountSim | Java **Jan 2025 - Feb 2025**

- Designed and implemented a bank simulator using Object-Oriented design in Java leveraging classes to securely store account information and transactions resulting in a robust banking simulator
- Applied the software development cycle to test and improve storage usage and runtime using Object-Oriented design resulting in a 30% improvement in runtime and storage usage
- Designed and implemented various algorithms to effectively validate user information enhancing the security and reliability of a banking simulator

Budget_Manager | Python **Dec 2024 - Jan 2025**

- Engineered a personal budget manager in Python, enabling users to add/remove expenses, priorities expenses, and exports simulating a real world financial management tool
- Utilized various data structures and algorithms including tuples, compacted lists, and reusable functions to optimize storing users' budget and expenses resulting in 20% reduction in storage usage
- Applied input validation with regex to enforce correct monetary values, strengthening error handling and improving reliability of the financial tool